

Curriculum Vitae - Ayush Agrawal

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Education

May 2022	Birla Institute of Technology and Science (BITS) Pilani	Pilani, India
Aug 2018	B.E. Electronics & Instrumentation GPA: 7.66/10	

Publications

J=Journal, C=Conference

- [J.1] **Physical Reasoning and Object Planning for Household Embodied Agents**  
[Ayush Agrawal](#), Raghav Prabhakar, Anirudh Goyal, Dianbo Liu
Transactions on Machine Learning Research [TMLR 2024]
- [C.1] **Sequence Agnostic Multi-Object Navigation**  
Nandiraju Gireesh*, [Ayush Agrawal](#)*, Ahana Datta*, Snehasis Banerjee, Mohan Sridharan, Brojeshwar Bhowmick, Madhava Krishna (* = Equal Contribution)
IEEE International Conference On Robotics And Automation [ICRA 2023]
- [C.2] **CLIPGraphs: Multimodal Graph Networks to Infer Object-Room Affinities**   
[Ayush Agrawal](#)*, Raghav Arora*, Ahana Datta, Snehasis Banerjee, Brojeshwar Bhowmick, Krishna Murthy Jatavallabhula, Mohan Sridharan, Madhava Krishna (* = Equal Contribution)
IEEE International Conference On Robot And Human Interactive Communication [RO-MAN 2023]

Experience

Feb 2024	National University Of Singapore (NUS)	Remote / Singapore
June 2023	Research Intern / Advisors: Dr. Dianbo Liu , Dr. Anirudh Goyal Developed a framework and QA datasets to analyse and improve physical commonsense reasoning in embodied agents.	
June 2023	International Institute Of Information Technology (IIIT) Robotics Research Center	Hyderabad
May 2022	Research Assistant / Advisors: Dr. K. Madhava Krishna , Dr. Mohan Sridharan , Dr. Krishna Murthy Developed computational methods inspired by human cognition to enhance the performance of embodied agents in object navigation, multi-object navigation, and household tidying-up tasks	
May 2022	University of New South Wales (UNSW) Bio-Engineering Lab	Remote / Canberra, Australia
Nov 2021	Research Intern (Bachelor Thesis) / Advisor: Dr. Sridhar Ravi Designed a deep learning model to mimic honey bee vision for obstacle avoidance in drones	
Aug 2021	Indian Institute Of Technology (IIT) ARMS Lab	Remote/ Mumbai, India
June 2021	Summer Intern / Advisor: Dr. Arpita Sinha Developed a Decentralized Multi-Drone Terrain Exploration algorithm using PX4 drones on ROS and Gazebo.	

Select Research Projects

- Physical Common Sense Reasoning** June'23 - Feb'24
Advisors: [Dr. Dianbo Liu](#), [Dr. Anirudh Goyal](#)
- Formulated a 3-step architecture to delineate physical commonsense reasoning for household tasks' object selection.
 - Created human preference datasets and 4 QA datasets to analyze physical commonsense reasoning capabilities of LLMs.
 - Finetuned LLMs on our datasets and analyzed the performance gain.[TMLR'24]
- Embodied Multi-Object Navigation** May'22 - June'23
Advisors: [Dr. K Madhava Krishna](#), [Dr. Mohan Sridharan](#), [Dr. Krishna Murthy](#)
- Developed commonsense-oriented heuristics to optimize the search and retrieval task of multiple objects by framing the problem as a Contextual TSP.
 - For static objects, developed a modular framework with an RL policy based on semantic inputs to output effective long-term goals thus enabling the robot to locate the list of objects in an optimized sequence agnostic manner[ICRA'23][Blog]
 - For dynamic objects, developed a Graph Neural Network by processing Human Preference Dataset and CLIP Features to give better human commonsense aligned Object-Room Affinities and latent embeddings.[RO-MAN'23][Intuition]

Bio-Inspired Robotics

Sept'21 - May'22

Advisors: [Dr. Sridhar Ravi](#), [Dr. Puneet Mishra](#), [Dr. Sujan Yenuganti](#)

- Developed a Deep Neural Network(LSTM+CNN) to model the relationship between Geometric Optic Flow and Honey Bee trajectories. Our developed obstacle avoidance algorithm achieved an accuracy of **75%** and was verified in various simulated multi-obstacle tunnels. [[Thesis](#)][[Presentation](#)]
- Proposed and Procured institute funding for developing Autonomous Source Localization Drone on a DJI Tello Drone utilizing the biologically inspired Run-Tumble Algorithm[[Report](#)]

Robotics & Electronics

Jan'21 - Jan'22

Advisors: [Dr. Arpita Sinha](#), [Dr. Meetha Shenoy](#), [Dr. Puneet Mishra](#),

- Implemented a Decentralized Multi-Drone Terrain Exploration algorithm on ROS and Gazebo using PX4 drones, ensuring complete exploration in a limited number of steps [[Code](#)][[Video](#)]
- Constructed a specialized dataset featuring common UAV noises and implemented a Deep Neural Network(GAN) for effective denoising of UAV-captured images [[Report](#)]
- Designed ESP32-AWS architecture for soil condition sensing by integrating microcontrollers and sensors to enable regular data transmission to AWS Database[[Presentation](#)]

Patents

Method And System For Multi-Object Tracking And Navigation Without Pre-Sequencing, 2023 | Patent Pending

Talks

“Bio-Mimicry”

- A Honey Bee's Attempt at Obstacle Avoidance [[🔗](#)]

December 2021 (BITS Pilani)

Honours and Awards

OpenAI Researcher Access Program, 2023 [[🌐](#)] For studying Physical Commonsense Reasoning abilities in LLMs

BITS Pilani Undergraduate Project Funding, 2021 [[🌐](#)] For working on Autonomous Odor Localization Drone

Bronze Medal, Univeristy Physics Competition 2021 [[🌐](#)] For presenting our solution as a white paper in 48 hours

Competitions

E-Yantra Robotics Competition Top 30 out of 500 teams [[🏆](#)]

Sep'20 - Apr'21

- Designed and simulated a parcel distribution drone featuring a Discrete PID Controller.
- Successfully executed marker detection, QR code-based scanning, path planning, obstacle avoidance, and optimized delivery/pickup scheduling to maximize revenue.

International Rover Design Challenge Core Member [[🏆](#)]

Jul'21 - Sep'21

- Contributed to implementing various low-light imaging techniques using Deep Learning and Image Processing techniques for our institute's submission.

Skills

Languages	Python, C++, MATLAB
Frameworks	Pytorch, Tensorflow
Tools	Git, Visual Studio
Simulators	Habitat, AI2Thor, Virtual Home, Gibson, Gazebo
Relevant Coursework	Pattern Recognition, Probability & Statistics , Linear Algebra, Discrete Mathematics Neural Networks & Fuzzy Logic, Object Oriented Programming, Digital Image Processing
MOOC	RL by David Silver, Deep Learning Specialization by Andrew Ng

Academic Service

Reviewer IROS 2023

References

- Dr. Dianbo Liu Assistant Professor, NUS, Singapore [[🌐](#)]
- Dr. K Madhava Krishna Professor, IIIT Hyderabad, India [[🌐](#)]
- Dr. Mohan Sridharan Reader, University of Birmingham, UK [[🌐](#)]
- Dr. Krishna Murthy Jatavallabhula PostDoc, MIT [[🌐](#)]